PROJECT OVERVIEW

ECHOBRIDGE HACKATHON

An Enterprise Network Design Challenge

[VLANs] [Sub-interfaces] [Packet Tracer]

CS WITH HUZAIFA

NETSEC
[NETWORKINS] TEAM
BATCH 2

01 / AGENDA

1. The Client Brief

Unpack business requirements and financial constraints.

2. The Blueprint Translation

Convert business logic into technical architecture.

!

3. The 60-Minute Build

Execute the deployment in Packet Tracer/GNS-3.

4. The Final Pitch

Defend the design to non-technical stakeholders.

THE EXPECTATION

“I need flawless, low-latency voice calls and central database access for 24 agents and 1 manager.”

THE CONSTRAINT

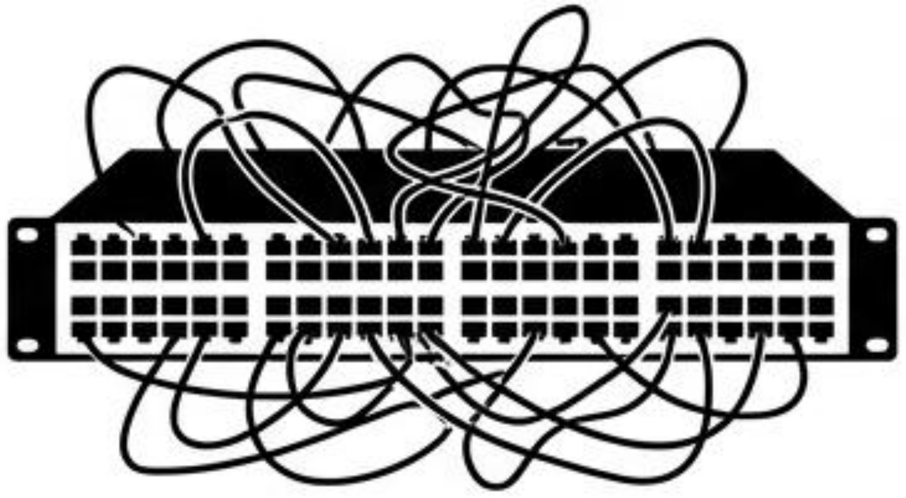
Strict 1.5 Million PKR budget. Do not go over budget.

NON-TECHNICAL CLIENTS SPEAK IN BUDGETS, NOT PROTOCOLS.

03 / THE HACK

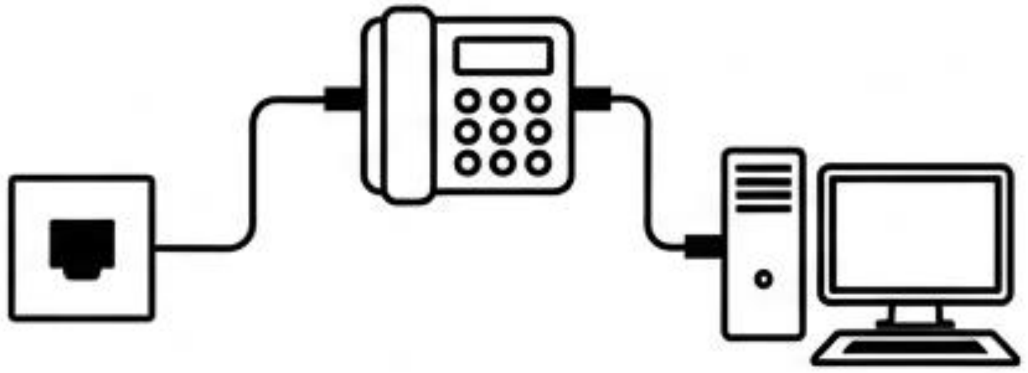
Daisy-Chaining Saves the Budget

THE TRAP



- 50 separate physical cable runs
- Requires massive 48-port PoE switch
- **Result: BLOWS THE 1.5M BUDGET.**

THE HACK



- 25 cable runs
- Daisy-chain PC into IP Phone
- **Result: UNDER BUDGET.**

THE TECHNICAL CHALLENGE:

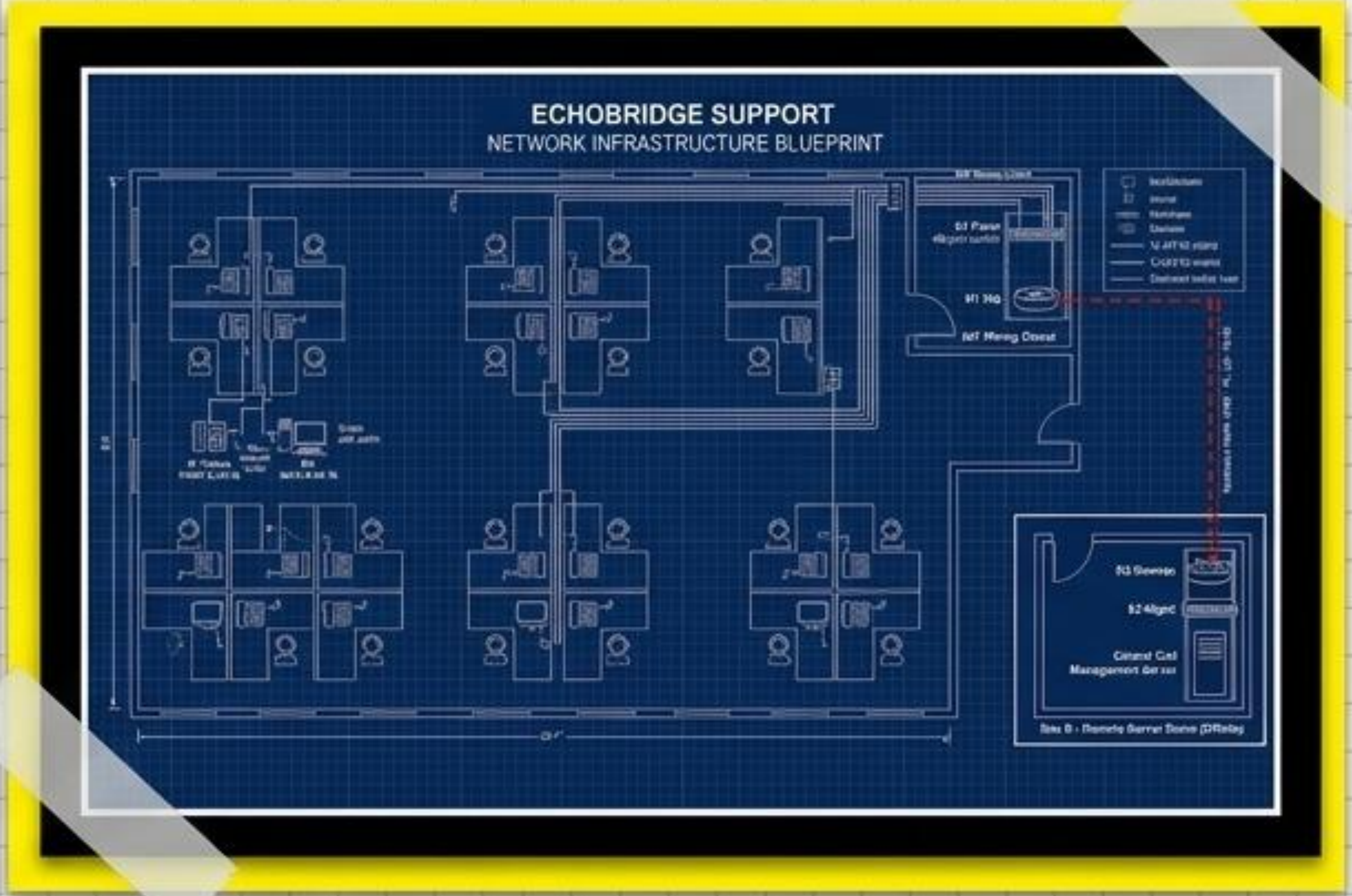
How do we separate heavy PC data traffic from sensitive Voice traffic on a single switchport?

04 / THE STACK

Network Subnets & Routing Rules

Technical Stack

VLAN 10	**Data** | Subnet: 192.168.10.0 /24 Gateway: 192.168.10.1 (Router 1)
VLAN 20	**Voice** | Subnet: 192.168.20.0 /24 Gateway: 192.168.20.1 (Router 1)
VLAN 30	**Servers** | Subnet: 10.1.1.0 /24 Gateway: 10.1.1.1 (Router 2)



REQUIRED INFRASTRUCTURE: Router-on-a-stick (802.1Q) for Inter-VLAN Routing + Static Routes across the 172.16.1.0/30 WAN

05 / EXECUTION

60 Minutes on the Clock.

60:00

[BUILD]

Construct the physical topology in Packet Tracer/GNS-3 based on the blueprint.

[CONFIG]

Implement CLI: Set up trunk/access ports, Voice VLANs, sub-interfaces, and static routes.

[VERIFY]

Ping the central server (VLAN 30) from the agent PC (VLAN 10). Verify Voice VLAN status on switchports.

Defend Your Design

- ✔ You must present your working network to the Business Owner.
- ✔ Explain how Voice VLANs guarantee calls won't drop.
- ✔ Explain how daisy-chaining saved the 1.5M PKR budget.

WARNING: NO TECHNICAL JARGON ALLOWED. Explain WHY it solves the business problem, not HOW you typed the commands.

Quiz

SCAN IT

You must present your working network to the Business Owner.

Explain how Voice VLANs guarantee calls won't drop.

Explain how daisy-chaining saved the 1.5M PKR budget.

WARNING: NO TECHNICAL JARGON ALLOWED. Explain WHY it solves the business problem, not HOW you typed the commands.